TrES-3b

R-1467

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_190518` · created 2026-08-01T11:27:53+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458621.7205 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.053
Transit depth (Rp/Rs)^2	3.82 +/- 2.08 [%]
Orbital Inclination (inc)	81.34 +/- 2.23
Airmass coefficient 1 (a1)	1.057 +/- 0.033
Airmass coefficient 2 (a2)	-0.0308 +/- 0.0086
Scatter in the residuals of the lightcurve fit is	3.04 %
Best Comparison Star	#2 - [25, 89]
Optimal Method	PSF photometry
Transit Duration (day)	0.047 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.053 vs 0.16309 (deviation 0.6σ)
Duration fit vs published	0.047 d vs 0.0567 d (deviation 0.39σ)
Mid-time O–C	0.3 min
Depth SNR	3.7
Residual scatter	3.04 %

Light curve

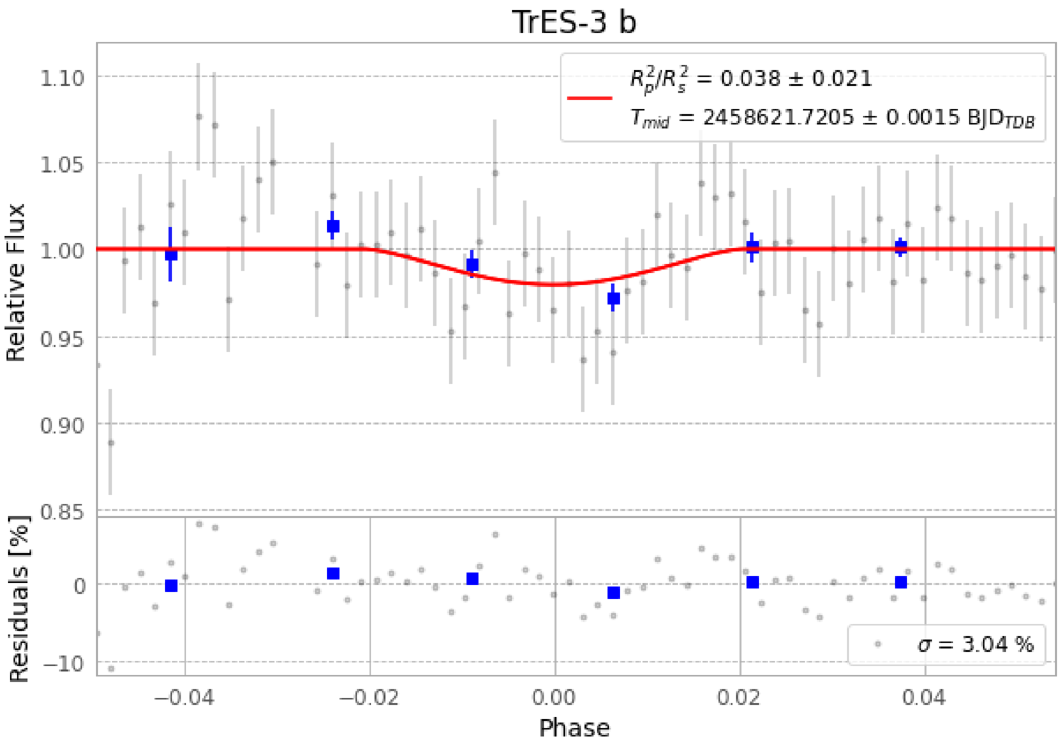

Figure 1 — FinalLightCurve_TrES-3 b_2019-05-17.png (SHA-256 a7df8b59b57c9824…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 43], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 694df93c6d5224ce…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

67 FITS frames (44 MB), `TRES-3190518033612.FITS` → `TRES-3190518065412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190518.log (8c240c3eed18…), FinalLightCurve_TrES-3 b_2019-05-17.png (a7df8b59b57c…), FinalLightCurve_TrES-3 b_2019-05-17.csv (64c295c968d2…)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)